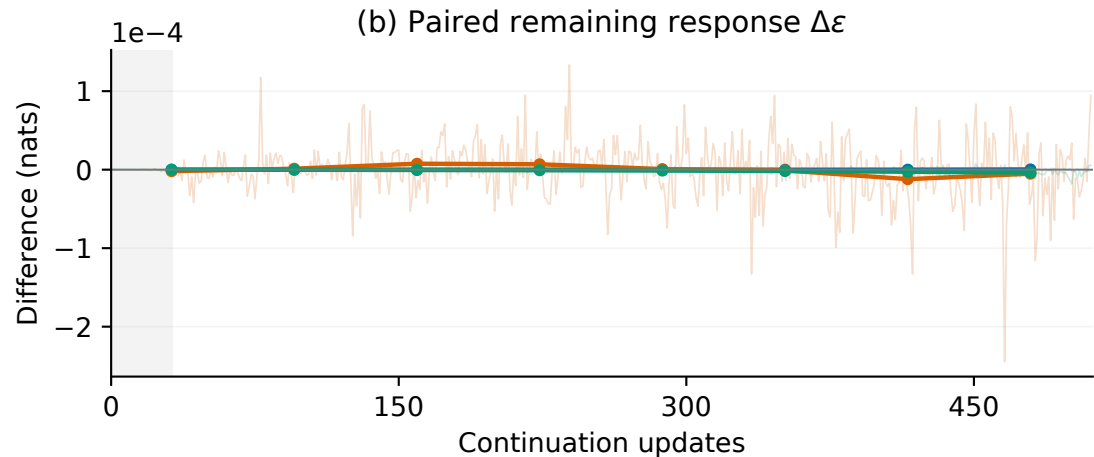


—●— SGD – Heavy Ball effective  $h=3e-08$ 
—●— SGD – Heavy Ball effective  $h=3e-07$ 
—●— Adam – Head only (different settings) raw  $\eta=3.03e-09 / 3e-08$



Light lines: observed values. Dots: 64-step means. Panel (a): solid  $\Delta L$ ; dashed  $\Delta A + \Delta Q_2$ .  
 Common difficulty cancels. Original shared step axis; no fitted time alignment.  
 Terminal local-scale protocol (adaptive).